

A portfolio of my previous projects

- *I've always enjoyed taking on projects that push me to think creatively. This portfolio showcases some of my favorite achievements, which I believe highlight the enthusiasm and passion I will bring to your team if given the opportunity*
- *Aditya Goshalia*

Project 1: Aerial Seeding Tank to be attached to a 42 lb drone

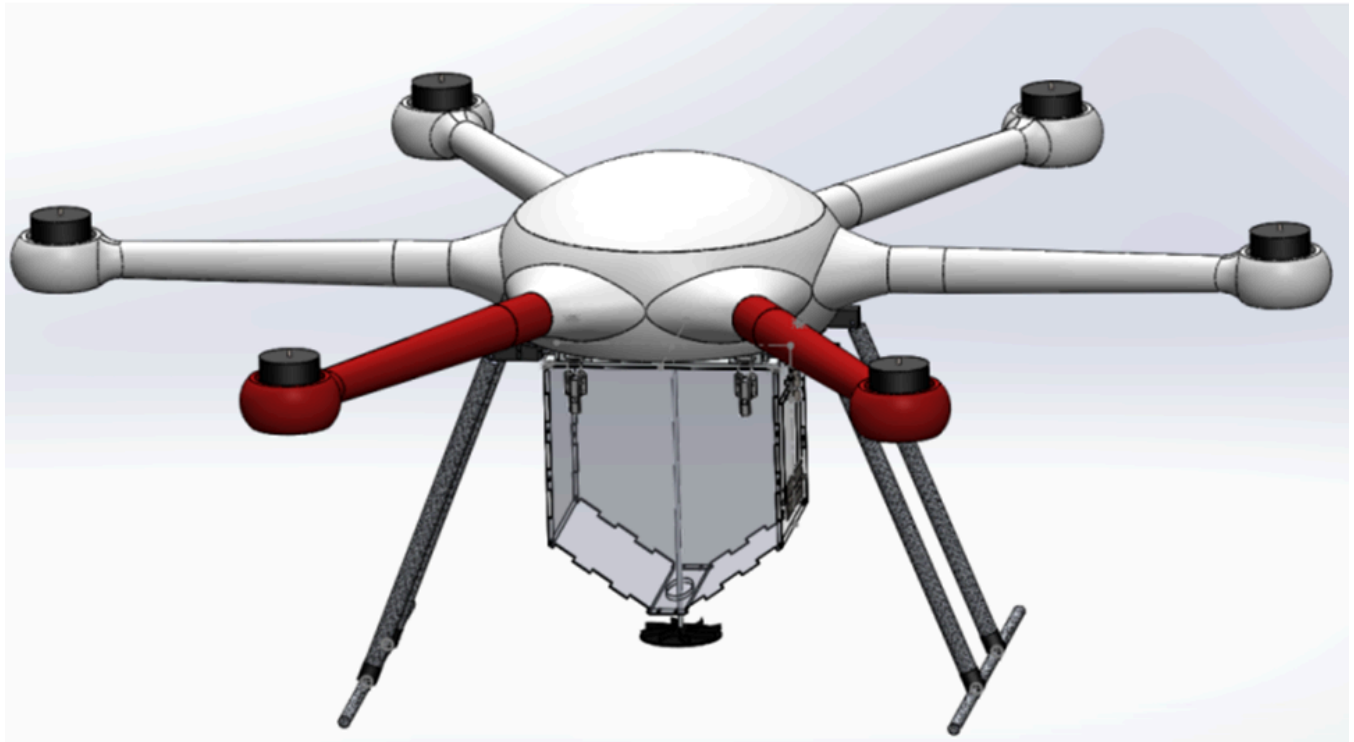


Figure 1.0: Final CAD Design of tank mounted to the drone.

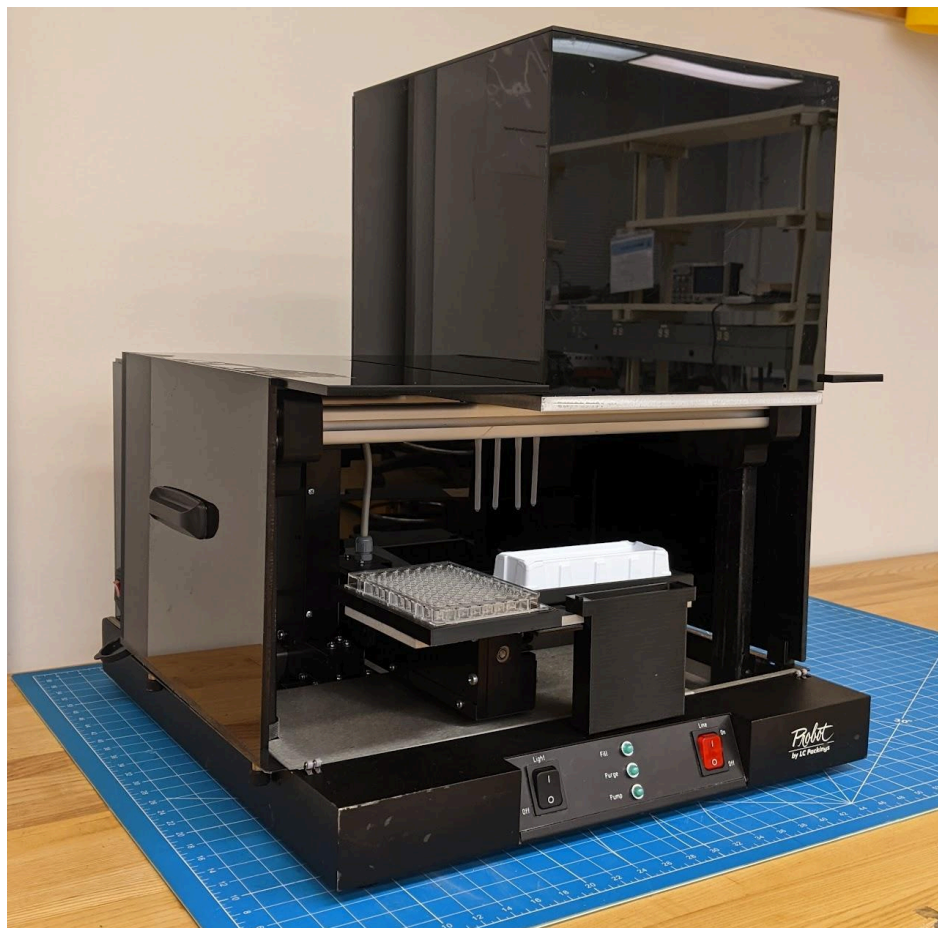
Description:

The inspiration behind this project came from a hiring fair at UBC where I learned about Stinson Aerial Services, a startup in Courtenay, BC, which had a vision to revolutionize agricultural practices by aerially seeding vast acres of land. This initiative aimed to ease the burden on hardworking indigenous farmers. Partnered with a UBC Mechanical Engineering student, I embarked on an extensive design process that involved multiple reviews and close collaboration with on-campus machine shops. The result was a fully functional prototype, poised for testing this fall by the company. The system includes 8 polycarbonate plates sealed together via silicone sealant. Draw latches attach the top plate to the rest of the tank for easy removal. A user fills seed in through a side door. Once the seed is full the door closes and the tank is hooked up to a drone. A DC motor control spins a shaft which goes through the tank and attaches to an impeller at the end which scatters seeds in a circular fashion during flight.



Figure 1.1: Final Tank Product Manufactured. To be shipped to client in Nicola Valley, BC for test flights

Project 2: Automated Hydrogel and Dispensing System



*Figure 2.0 Automated Hydrogel Dispensing System Final Finished Product
UBC Biomedical Engineering Capstone - First Place Award Winner*

Description:

Probably the most rewarding project I have worked on. The automated hydrogel dispensing system was a year-long capstone R&D project which allowed myself and my team to automate the dispensing/aspiration procedure usually done by lab technicians. The system consisted of a stage and 96 well plate which were bolted to a moving XYZ motorized drive. 4 custom made nozzles each protected by IP and an NDA for our client were developed and 3D printed by myself and my team which allowed precise aspiration and dispensation of reagents. We custom

designed a reservoir for larger volumes of up to 25 mL and a mini-tube rack for smaller volumes as low as 20uL. I am proud to say after a year of hard work we were able to reach $\pm 5\mu\text{L}$ ballpark precision in the dispensing/aspirating area with our nozzles. Finally, we built a GUI called FluidCAM that allowed users to generate and run procedures on the machine without the need to understand gcode or python programming languages. The user is then able to input the volumes they desire to be dispensed into each of the wells in the plate using the 96 well grid.

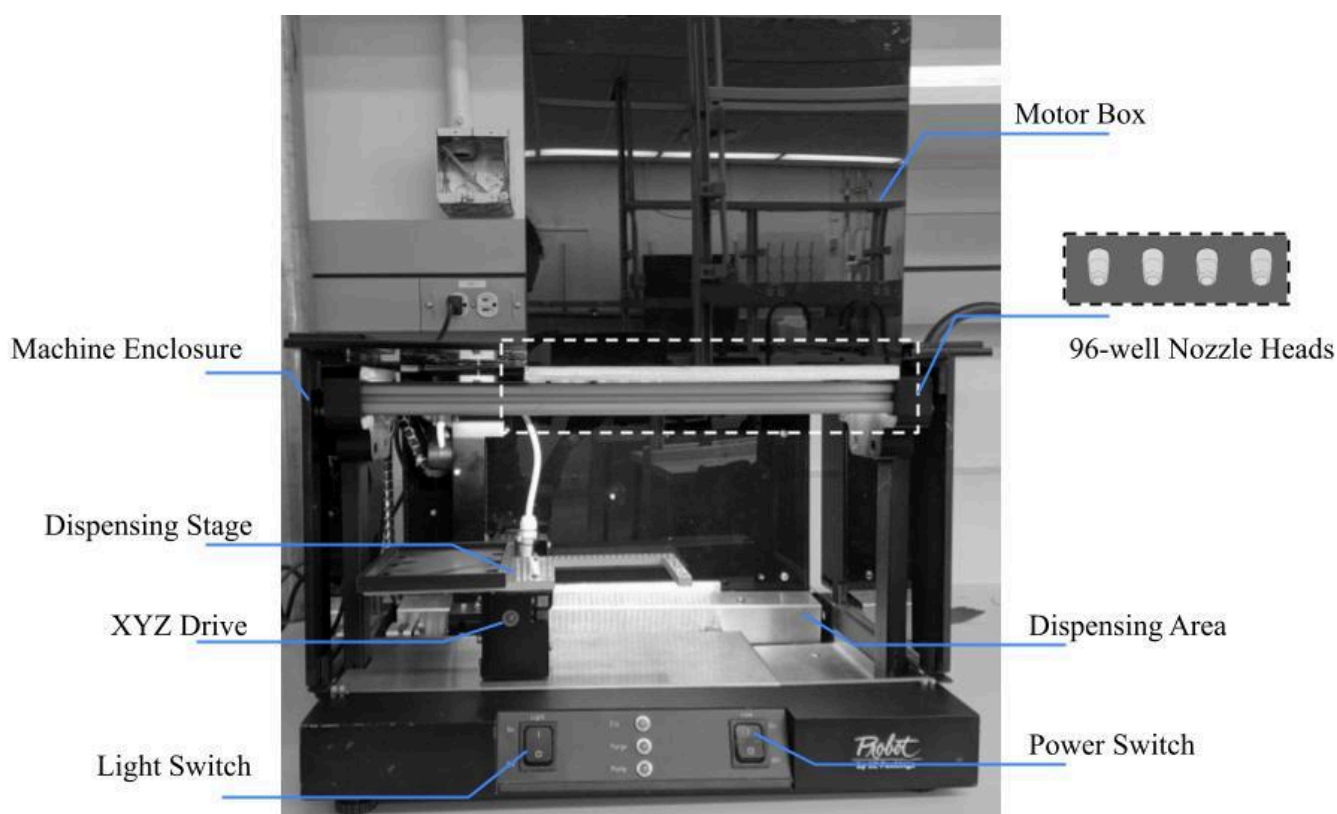


Figure 2.1 Front View of Hydrogel Dispenser - Stage in Home Position

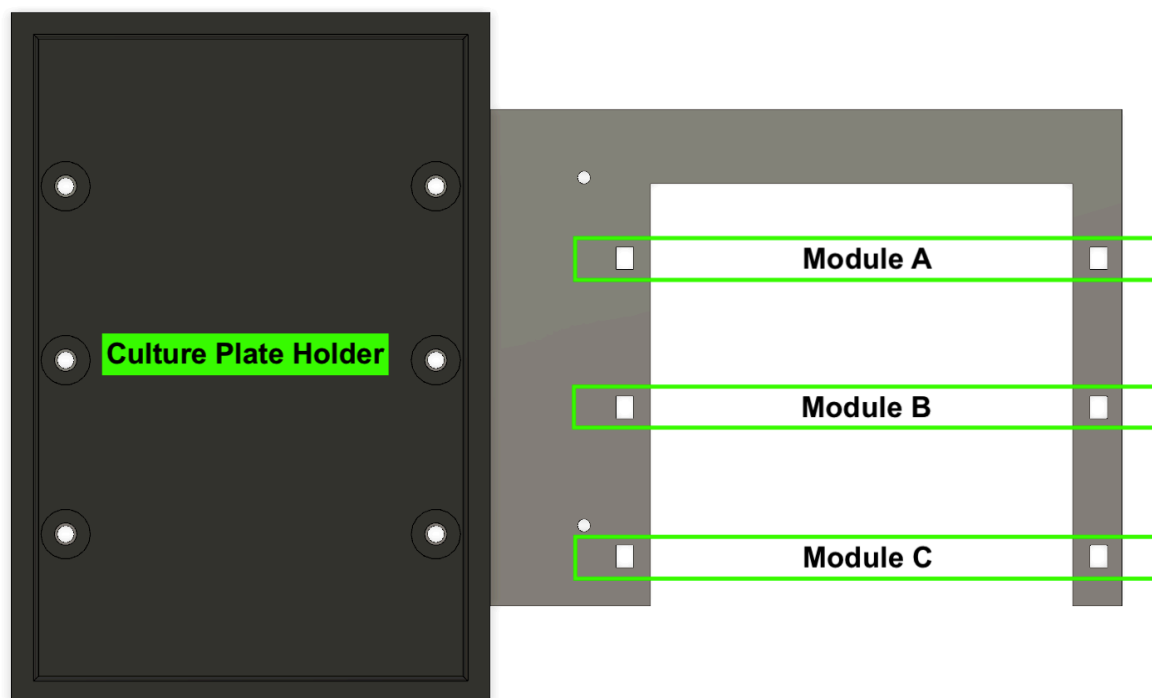


Figure 2.2 : Diagram of Dispensing Stage Layout.

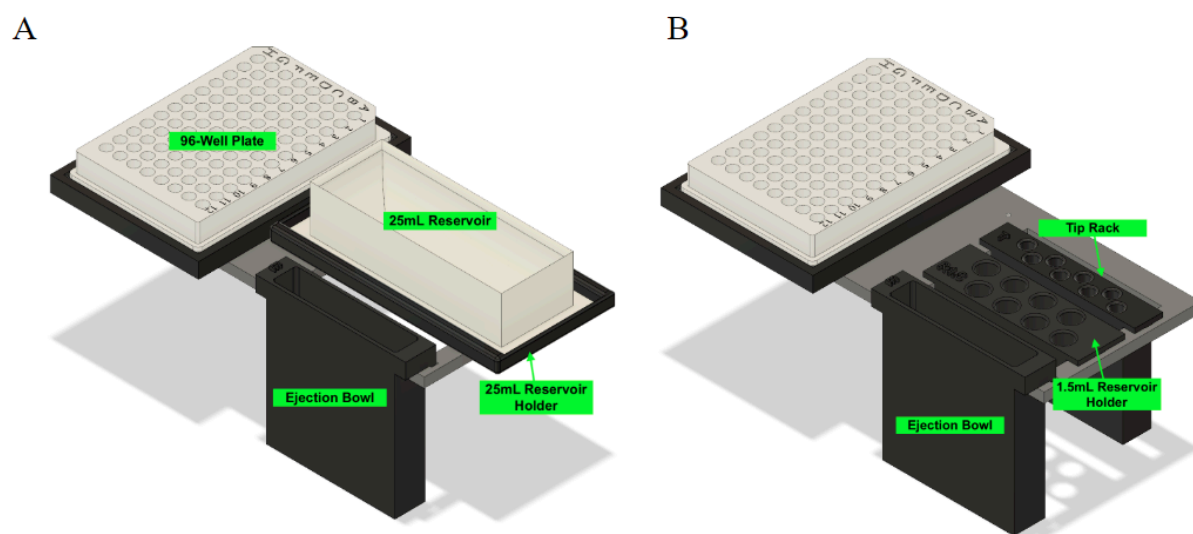


Figure 2.3 : Dispensing stage low and high volume set ups. A) High volume set up for 25 mL reagent reservoir. B) Low volume set up for 1.5 mL reagent reservoirs.

FluidCAM

Enter procedure name:

☐ Set All
 ☐ A
 ☐ B
 ☐ C
 ☐ D
 ☐ E
 ☐ F
 ☐ G
 ☐ H

Plate Size: {Choose a plate size}
 Insert Type: {Choose an insert type}
 Reservoir Type (in mL): {Choose a reservoir type}
 Tip Type (in uL): {Choose a tip type}
☐ Auto-Equip Tips
☐ Auto-Eject Tips

☐	1								
☐	2								
☐	3								
☐	4								
☐	5								
☐	6								
☐	7								
☐	8								
☐	9								
☐	10								
☐	11								
☐	12								

Enter volume (in uL) to fill the grid:

Tip Placement
A B C D

Reservoir Volumes
Generate Procedure To View Volumes

Generate Procedure

Exit

Import CSV File Load Preferences Save Preferences

Figure 2.4 : Custom GUI for generating dispensing procedures.

Project 3: Anodic Bonding Station for annealing microfluidic devices

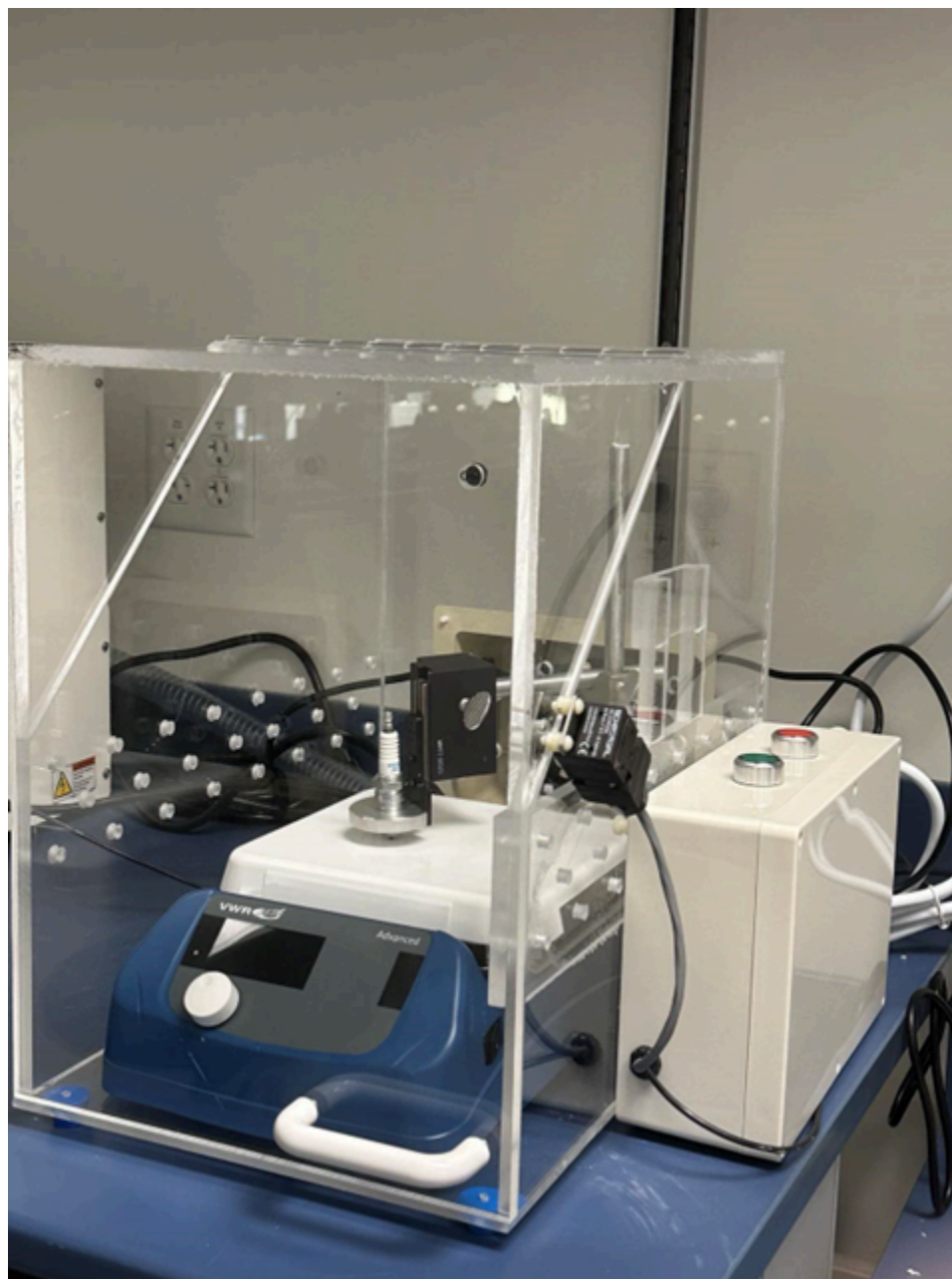


Figure 3.0: Polycarbonate Enclosure around a hot plate. Paired with a master control switch and connected to an external high voltage power supply (not shown in diagram).

Description:

As a research assistant co-op at a Biomedical Engineering Lab, my PI assigned me the task of building a cost-effective instrument to seal microfluidic devices under high heat and voltage, as commercially available options were prohibitively expensive. For this project, I utilized a hot plate from the lab as the heat source and designed a plexiglass enclosure with air vents to fit around it. I integrated a master switch connected to a safety latch on the enclosure door, ensuring that the box could not be opened while the system was active. To generate the necessary electric field, I purchased a spark plug from Home Depot and mounted it on a metal clamp, allowing it to serve as an electrode. The setup was powered by an external high-voltage power supply, which delivered 1200V through a wire connected to the spark plug, initiating the electric field required for the bonding process.

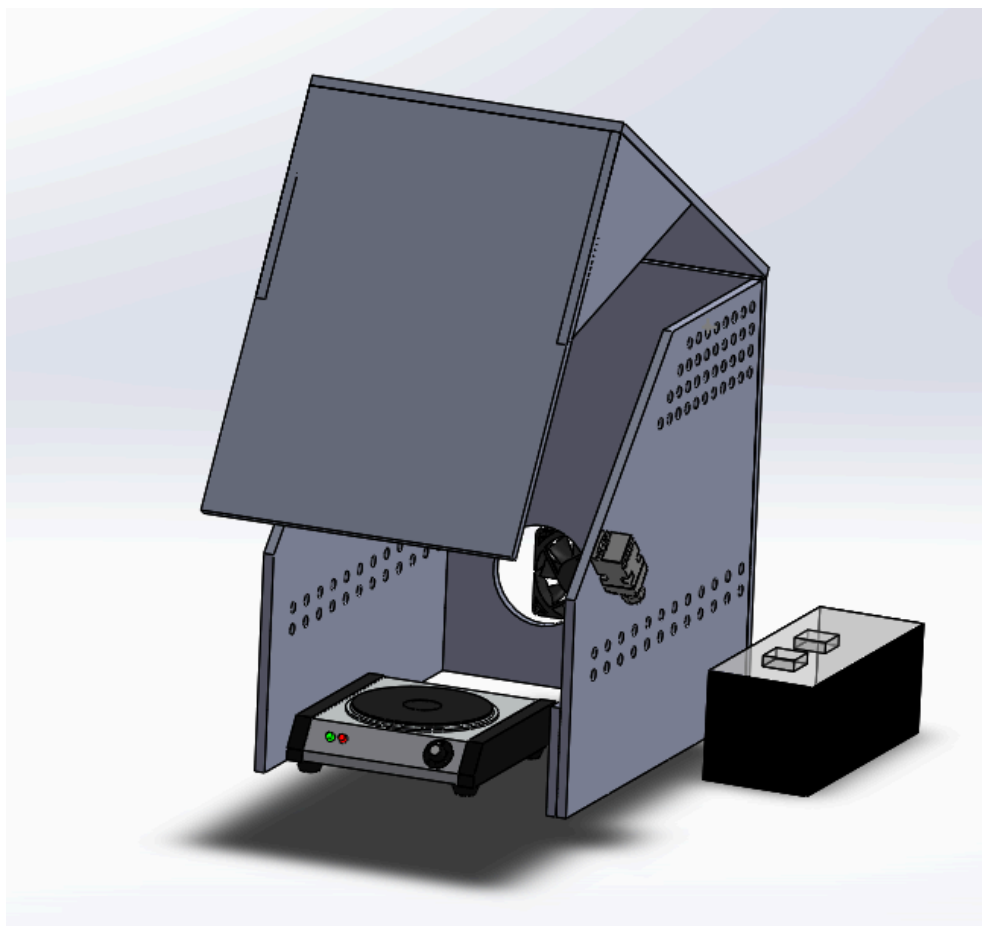


Figure 3.1: Front view - Opening and closing mechanism of the anodic bonding station

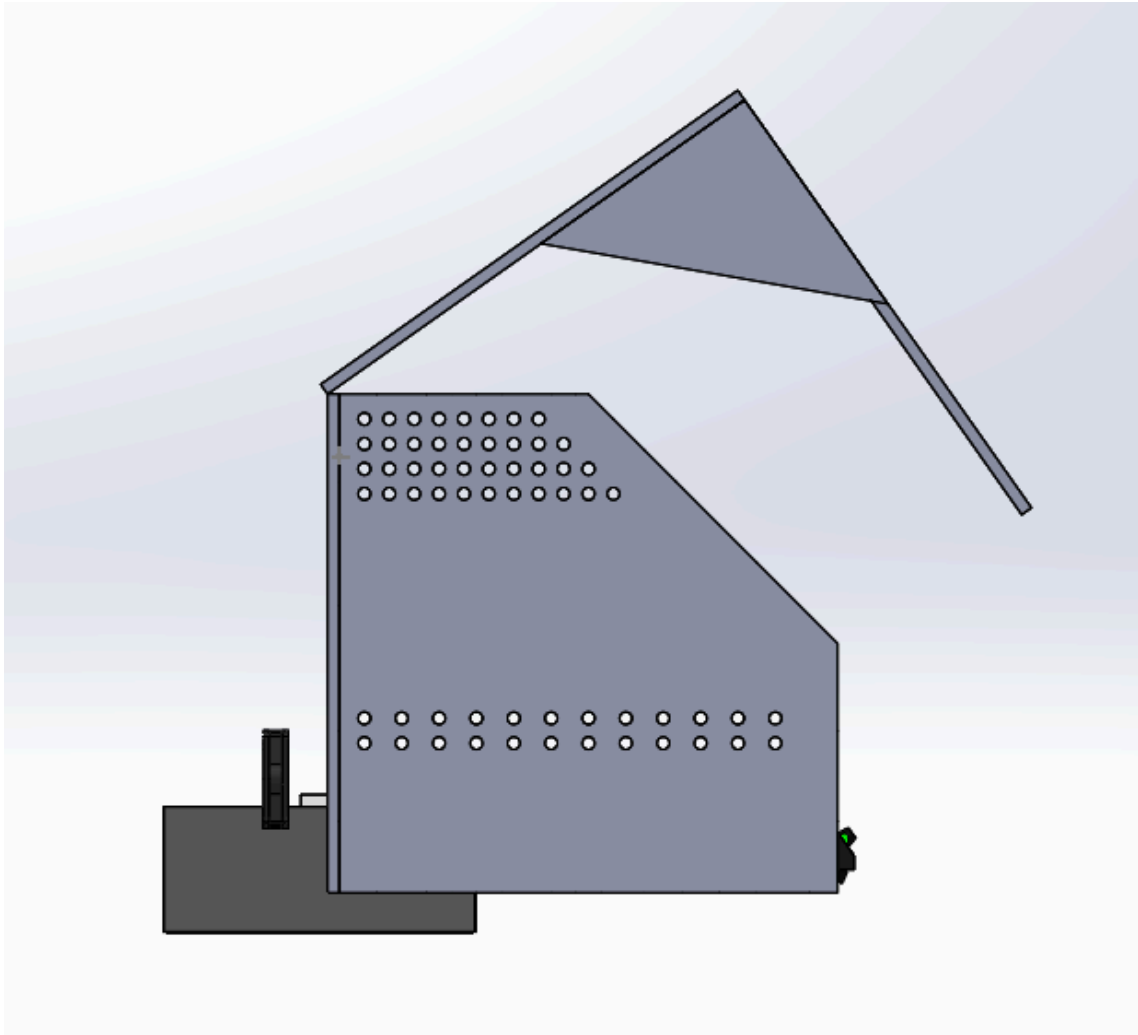


Figure 3.2: Side View of the Bonding Station - Air holes allow heat to dissipate as system warms up

Project 4: Environmental Chamber around a Nikon TI microscope for the purpose of creating an enclosed optimal environment for cell growth

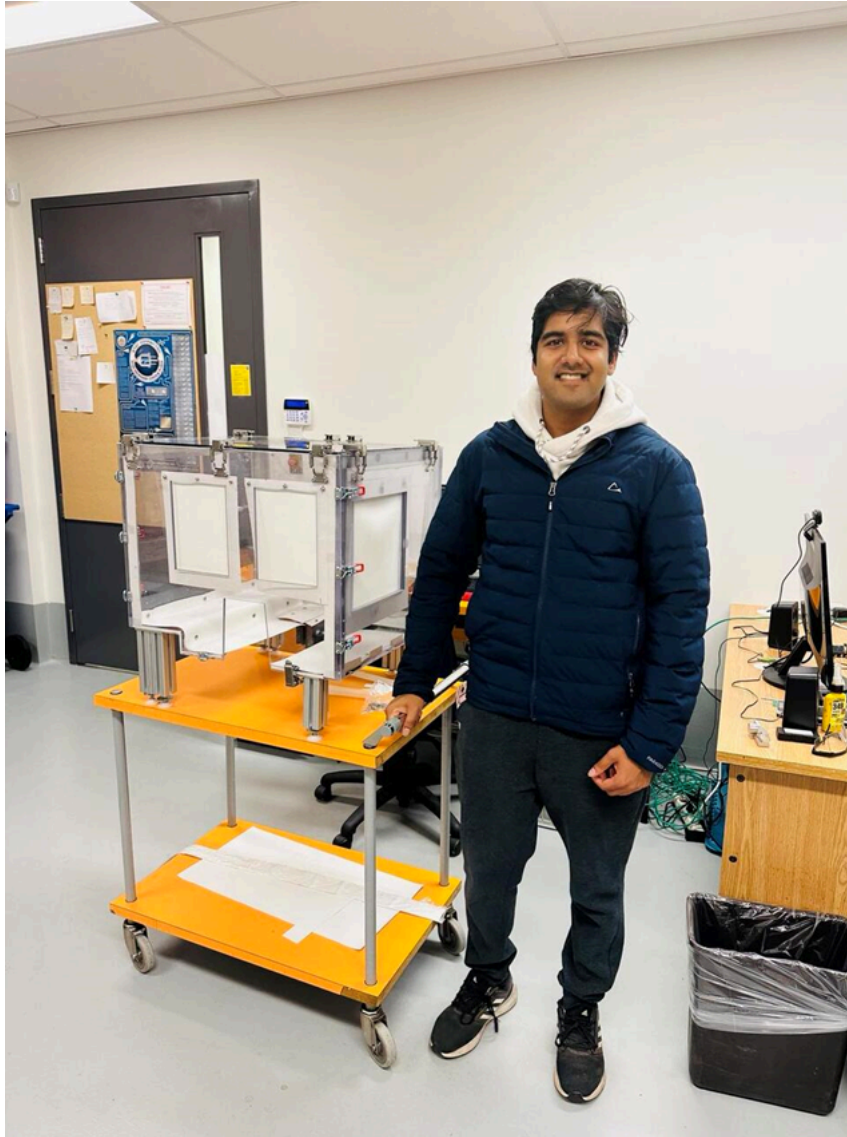


Figure 4.0: Environmental Chamber Finished Product.

Description

After nearly six months of dedicated effort, this project—building a custom environmental chamber for a Nikon-TI microscope—stands as one of my most rewarding accomplishments. The chamber was meticulously designed to create an

optimal environment for cell cultures to thrive while allowing researchers to observe their growth in real-time through the microscope. The structure featured latches on the sides and top, enabling easy assembly and disassembly by researchers. The chamber's stability was ensured by securing its eight legs into a vibration-proof table. To maintain the required conditions for cell growth, an air heater with a PID controller regulated the temperature at 37°C, while a CO₂ unit provided the necessary atmosphere for the cells.

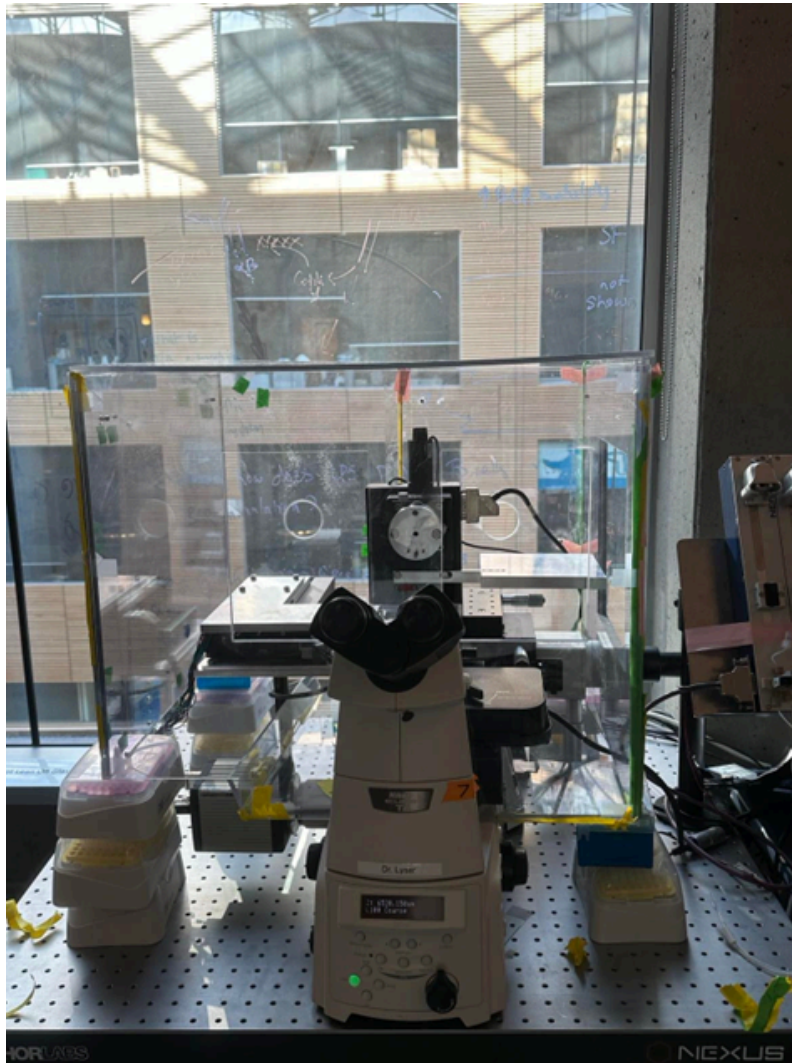


Figure 4.1: Early Prototype of chamber without legs and latches enclosed around microscope

Project 5 - PDMS Microfluidic Probe Heads

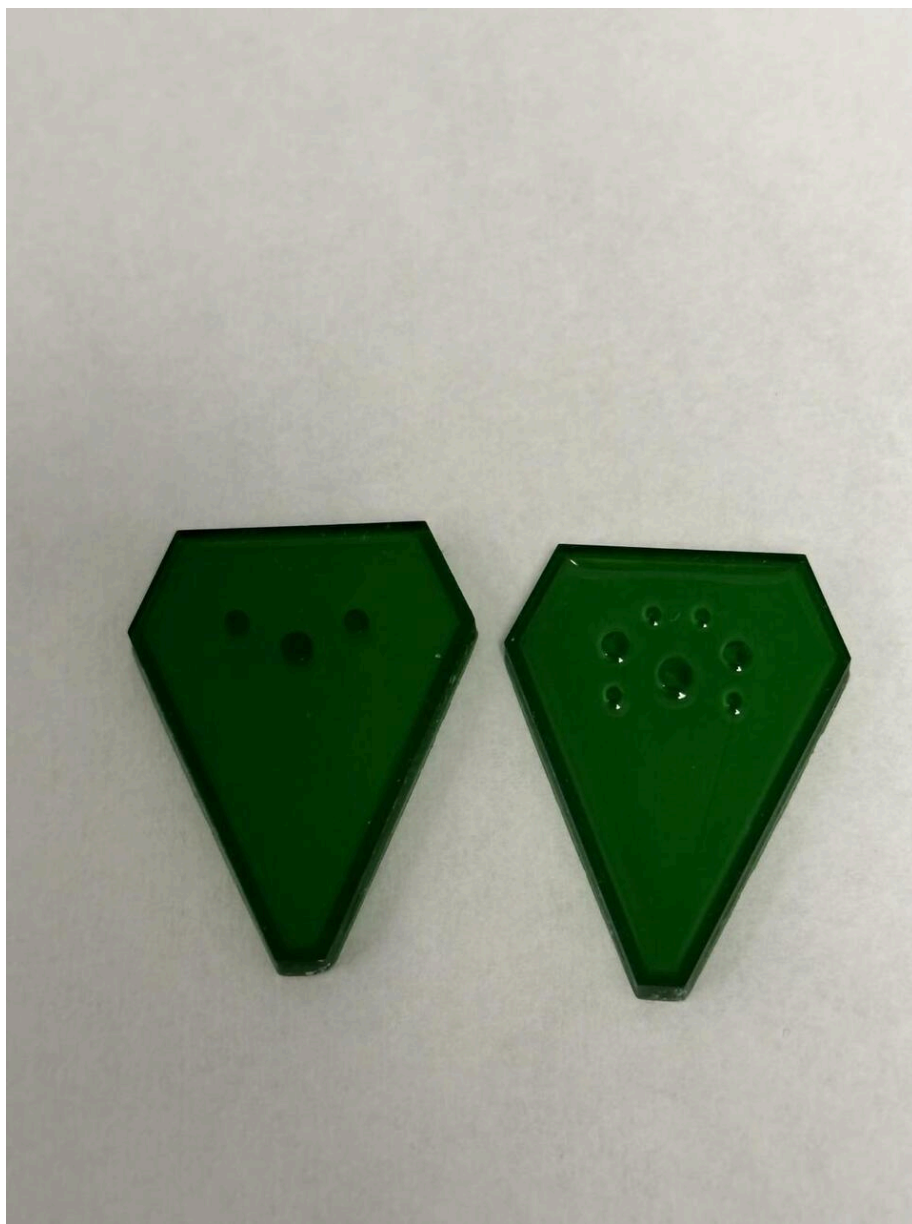


Figure 5.0: Microfluidic Probe Heads made using PDMS with microchannels.

Description

The project involved extensive use of microscopy and fluorescence imaging techniques to develop test microfluidic devices with microchannels, designed to deliver precise microvolumes of reagents or drugs to targeted surfaces like cancer cells or spheroids. I gained significant experience in microscopic fluorescence microscopy imaging techniques, as they were critical for

continually assessing the integrity of the microchannels to ensure they remained unobstructed and functional.

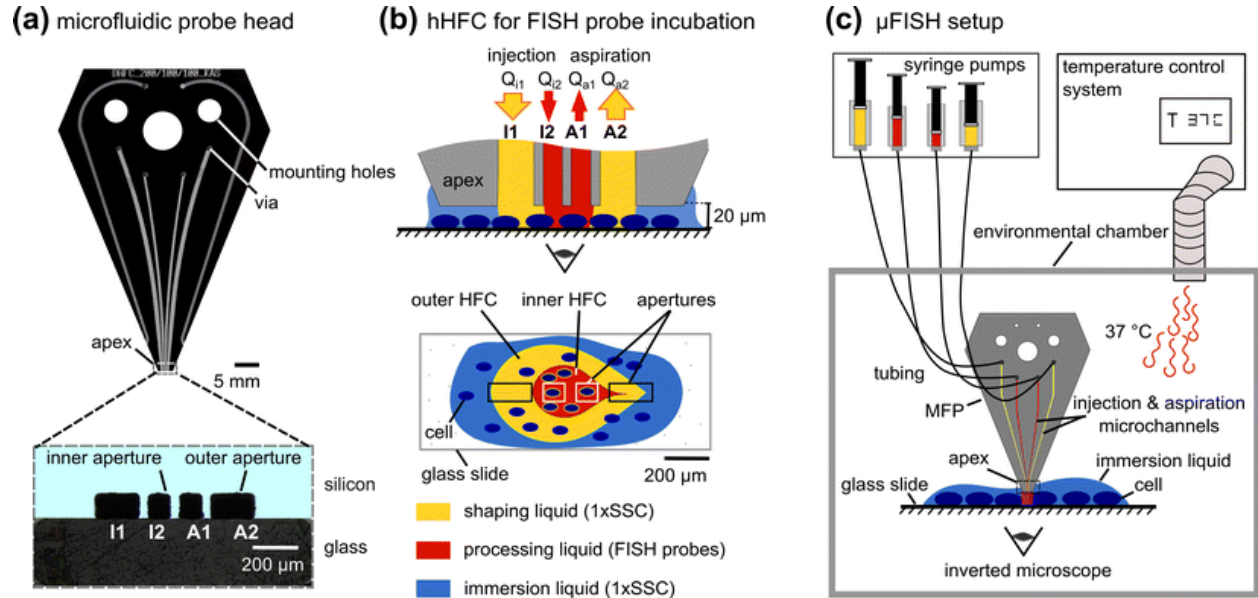


Figure 5.1: Microfluidic device consists of an injection and aspiration microchannel channel. Device is hooked up to the front of an inverted microscope with the tip of the device just above the substrate slide. Syringe pumps inject reagents/drugs through the device's channels onto a localized part of the substrate.

Reference - Huber, D., J. Autebert, and G. V. Kaigala. "Micro Fluorescence in Situ Hybridization (μFISH) for Spatially Multiplexed Analysis of a Cell Monolayer." Biomedical Microdevices, vol. 18, no. 2, 2016, pp. 40-40.